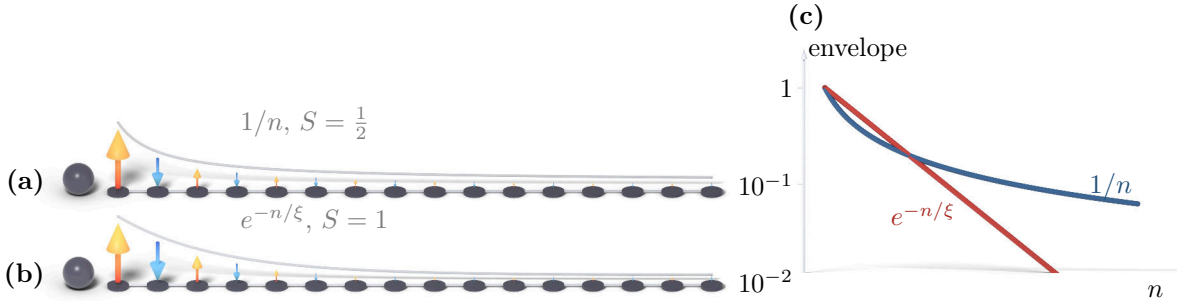

(a)
$1/n$, $S = \frac{1}{2}$
(b)
$e^{-n/\xi}$, $S = 1$
(c)
envelope
1
10^{-1}
10^{-2}
$1/n$
$e^{-n/\xi}$
n